

CHURAPCHA, Russian Federation

WMO: 247680

Lat: 62.033N

Lon: 132.600E

Elev: 609

StdP: 14.38

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -61.8	(c) -58.9	(d) -65.3	(e) 0.1	(f) -61.3	(g) -62.2	(h) 0.1	(i) -58.4	(j) 8.9	(k) -24.5	(l) 7.7	(m) -26.6	(n) 0.1	(o) 220	(p) 0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	86.5	65.2	82.9	63.8	79.1	62.7	68.1	80.6	66.3	77.9	64.3	75.5	5.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.2	91.9	71.9	61.9	84.8	71.5	59.8	78.4	69.6	32.9	80.4	31.4	78.1	29.9	75.5	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.8	9.9	-64.1	90.9	4.0	3.5	-67.0	93.4	-69.3	95.5	-71.6	97.4	-74.5	99.9
			-64.1	72.1	4.0	4.0	-67.0	74.9	-69.3	77.3	-71.6	79.5	-74.5	82.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	-40.6	-31.5	-5.6	22.9	45.2	60.4	65.8	58.8	42.0	16.4	-19.6	-40.4
	DBStd	40.43	10.99	11.14	13.66	11.27	7.66	7.43	6.92	7.01	7.95	12.80	16.41	11.58
	HDD50	14027	2809	2281	1724	812	188	9	1	10	260	1041	2089	2803
	HDD65	18509	3274	2701	2189	1262	614	173	76	217	690	1506	2539	3268
	CDD50	1155	0	0	0	0	39	322	491	283	20	0	0	0
	CDD65	161	0	0	0	0	0	36	101	24	0	0	0	0
	CDH74	2128	0	0	0	0	11	588	1240	287	2	0	0	0
	CDH80	649	0	0	0	0	0	154	431	64	0	0	0	0
	WSAvg	3.6	1.5	1.6	3.1	4.6	5.9	5.1	4.6	4.2	4.3	3.9	2.6	1.5
	PrecAvg	10.2	0.2	0.2	0.2	0.3	0.9	1.5	2.1	1.7	1.2	0.9	0.6	0.3
(15) Precipitation	PrecMax	15.6	0.7	0.6	0.6	0.9	2.2	3.5	4.7	3.5	2.6	2.0	1.5	0.7
	PrecMin	6.9	0.0	0.0	0.0	0.0	0.0	0.3	0.5	0.2	0.1	0.2	0.1	0.0
	PrecStd	2.0	0.2	0.2	0.2	0.2	0.5	0.9	1.1	0.9	0.6	0.4	0.3	0.2
	PrecStd	2.0	0.2	0.2	0.2	0.2	0.5	0.9	1.1	0.9	0.6	0.4	0.3	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.7	9.7	37.0	54.0	75.0	88.5	92.9	86.7	70.8	44.3	28.2	-1.6
		MCWB	-2.4	8.8	30.9	41.3	55.4	65.7	67.8	66.3	55.7	39.1	26.5	-2.2
	2%	DB	-12.8	-0.8	29.5	48.3	69.6	84.4	88.2	80.4	64.0	38.4	20.5	-12.6
		MCWB	-13.3	-1.4	24.8	37.8	52.0	63.2	66.4	64.5	52.4	34.0	19.3	-13.1
	5%	DB	-18.7	-8.1	23.8	44.1	65.3	80.8	84.9	76.9	59.4	35.0	11.7	-17.8
		MCWB	-19.0	-8.7	20.2	35.5	49.6	61.4	65.0	62.2	50.0	31.8	10.7	-18.2
	10%	DB	-24.6	-14.1	18.2	39.9	60.8	76.9	81.3	73.0	55.5	31.8	4.8	-23.8
		MCWB	-24.8	-14.5	15.5	32.7	47.4	59.7	64.1	60.2	48.8	29.7	4.1	-24.0
	0.4%	WB	-2.4	9.0	30.7	42.0	56.9	68.3	71.5	69.3	57.2	39.5	26.8	-2.3
		MCDB	-1.7	10.1	36.8	52.7	71.9	82.4	84.6	78.6	67.2	43.3	28.0	-1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-13.1	-1.7	24.9	38.5	53.4	65.3	69.1	66.5	54.5	34.6	19.2	-12.9
		MCDB	-12.9	-1.0	29.3	47.6	67.4	79.4	82.6	76.8	61.1	37.4	20.2	-12.6
	5%	WB	-19.0	-8.7	20.2	35.9	50.4	63.2	67.5	64.2	52.1	32.0	10.6	-18.2
		MCDB	-18.7	-8.1	23.6	43.1	63.0	77.0	79.8	74.1	57.6	34.4	11.6	-17.8
	10%	WB	-24.8	-14.6	15.4	33.2	48.2	61.3	65.9	61.8	49.4	29.7	4.1	-24.0
		MCDB	-24.5	-14.1	18.1	39.2	58.9	74.1	77.1	70.6	54.4	32.0	4.8	-23.8
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	21.6	30.7	24.5	19.9	22.0	21.0	20.4	17.3	14.4	14.6	10.6
		MCDBR	17.8	24.7	31.0	23.3	28.0	29.1	28.2	26.5	25.3	15.4	18.3	17.0
		MCWBR	17.5	24.0	26.7	15.2	13.9	11.7	10.3	12.0	14.5	11.3	17.6	16.9
	5% WB	MCDBR	18.0	24.6	30.1	21.9	25.1	26.6	23.1	22.1	20.1	13.7	17.8	17.2
		MCWBR	17.8	23.8	26.1	14.5	13.3	12.1	10.1	11.0	12.4	10.7	17.2	17.1
		MCWBR	17.8	23.8	26.1	14.5	13.3	12.1	10.1	11.0	12.4	10.7	17.2	17.1
(40) Clear-Sky Solar Irradiance	taub		0.207	0.255	0.299	0.359	0.442	0.406	0.488	0.435	0.346	0.284	0.215	0.178
	taud		1.979	2.081	2.066	2.030	2.042	2.290	2.030	2.160	2.382	2.285	1.989	1.889
	Ebn at Noon		152	223	252	258	244	259	230	233	239	213	142	100
	Edh at Noon		12	23	33	43	47	37	47	38	25	18	11	7
(44) All-Sky Solar Radiation	RadAvg		116	434	971	1562	1676	1870	1732	1337	811	375	159	54
	RadStd		5	18	36	60	142	135	100	99	87	36	12	3

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.35	+2.84	+2.16	N/A	N/A	N/A	-421	-472	+146	N/A
Regional (5 neighbors)		+1.28	+2.57	+2.27	N/A	N/A	N/A	-425	-443	N/A	N/A

Nomenclature: See separate page